Youngsuk Park

Email: youngsuk@cs.stanford.edu
Phone: +1 (650) 422-8541
Homepage: http://cs.stanford.edu/~youngsuk/
LinkedIn: https://www.linkedin.com/in/y-park/

Summary

Stanford PhD and AI research lead advancing algorithms for large-scale model training and inference. Lead the Core Algorithm team at AWS AI, developing efficient architecture design, low-precision training, and optimizers for AWS Trainium (Amazon's custom AI chip), in addition to agentic workflow and RL post-training alignment for optimizing DSL kernel synthesis. Solutions deployed across Amazon Bedrock, AGI, and Anthropic. Built the team from the ground up and currently manage 6+ scientists, presenting technical progress directly to VP/DE. Authored 36+ papers at ICML, NeurIPS, ICLR, and AISTATS. Active thought leader through 6 tutorials and workshop organizations, and 25+ PhD mentees placed at OpenAI, DeepMind, and Nvidia.

Research Focus: Efficient LLM Architecture | Low-Precision Training & Optimizer | RLVR & Post-Training Alignment | Agentic AI for Systems

Education

Ph.D. in Electrical Engineering, Stanford University, 2020
Advisors: Stephen P. Boyd and Jure Leskovec
Dissertation: Topics in Convex Optimization for Machine Learning

M.S. in Electrical Engineering, Stanford University, 2016

B.S. in Electrical Engineering (Minor in Mathematics), KAIST, 2013
Summa Cum Laude

Professional Experience (7+ years at Industry)

Principal Scientist & Sr. Research Manager, Amazon AGI Annapurna Science, Sept. 2026 – Present
Senior Applied Scientist & Research Lead, AWS AI, Mar. 2023– Aug. 2026

- Lead Core Algorithm team advancing scalable LLM modeling, training and inference for AWS Trainium
- Manage 6+ scientists (from Stanford, UCB, CMU) and 7+ PhD interns
- Develop Trn-optimized architectures for AGI-FMR pre-training team (by Chief Scientist Pieter Abbeel)
- Deploy efficient training and inference solution across Amazon Bedrock, AGI, and external customers
- Scaling RL system and reward design to optimize NKI & Triton DSL synthesis

Applied Scientist II, AWS AI Labs, Jun. 2020–Mar. 2023

- Technical lead for time series forecasting and foundation model development
- Lead third-party validation of foundation models on AWS Trainium accelerators

Research Intern, Adobe Research, Jun.–Sept. 2019

Research Intern, Criteo AI Lab, Jun.–Sept. 2018

Research Intern, Bosch Center for AI, Jun.–Sept. 2017

Tutorial, Workshop & Teaching

Tutorials:

- **KDD 2026**: Teaching LLMs to Write System Kernels for AI Accelerators: Post-Training, Reasoning, and Agentic Optimization (with R. Saha, J. Woo, Z. Jia)
- **AAAI 2026:** Algorithms and Systems for Efficient Inference in Generative AI (with R. Saha, Y. Wang)
- **IJCAI 2025:** Scaling LLM Training: Efficient Pre-training & Fine-tuning (with T. Yu, L. Lausen)

- **KDD 2024:** Inference Optimization of Foundation Models (with K. Budhathoki, J. Kübler, Y. Wang)
- **KDD 2023:** Training Foundation Models on Emerging AI Chips (with A. Muhamed, J. Huan)
- **IEEE Big Data 2022:** Deep Time Series Forecasting (with G. Gupta, J. Huan)

Workshop Organization:

- **Amazon Research Day 2026:** The Next Frontier of AI and Systems Co-design (Keynote by Ion Stoica)
- **KDD 2025:** Workshop on Inference Optimization for Generative AI, Toronto, Canada
- **KDD 2022:** Workshop on Mining and Learning from Time Series – Deep Forecasting, Washington DC

Stanford University:

- Teaching Assistant: Convex Optimization I (Instructor: Stephen Boyd), 2018
- Teaching Assistant: Convex Optimization II (Instructor: John Duchi), 2017

Publications

Preprint/Under Review

[1] LLM SYS J. Kim, R. Saha, **Y. Park**†. **Scaling Laws in Mixed Precision: Hardware-aware Compute Optimal.** Under review at ICLR.

[2] OPT LLM S. Kim, K. Ozkara, **Y. Park**†. **Optimizing LLM with Chained LMO: Muon and its Variants** Under review at ICLR.

[3] RL LLM SYS N. Sagan, R. Saha, J. Woo, **Y. Park**†. **Stabilizing 4-Bit RL: Loss-Level Corrections over Quantization-Aware Training.** Under review at ICLR.

[4] RL LLM Y. Li, J. Woo, **Y. Park**. **Multi-Reward Hacking: From Provable Collapse to Trust-Anchored Defense.** Under review at ICLR.

[5] LLM G. Wang, H. Lu, **Y. Park**, V. Kuleshov. **Revisiting Scaling Laws of Discrete Diffusion.** Under review at ICLR.

[6] RL LLM S. Wei, K. Ozkara, J. Woo, **Y. Park**. **Token-Level Teacher Selection for On-Policy Distillation.** Under review at ICLR.

[7] LLM SYS J. Yoo, R. Saha, S. Zhu, T. Yu, **Y. Park**†. **Modular Kernel Evolution: LLM-driven Kernel Synthesis for Domain Specific Language.** Under review at ML system, Appeared at ICML DL4C workshop.

[8] LLM SYS J. Huang, **Y. Park**, K. Budhathoki, J. Kübler, M. Kleindessner, Y. Wang, G. Karypis. **Accelerate LLM Inference via 4:8 Semi-structured Sparsity on AWS Trainium.** Under review at TMLR.

Peer-reviewed Publications

[9] LLM RL SYS J. Woo, S. Zhu, A. Nie, Y. Wang, **Y. Park**†. **TritonRL: Training LLMs to Think and Code Triton Without Cheating.** *Conference on Language Modeling (COLM)*, 2026.

[10] LLM RL W. Deng, J. Huang, K. Ozkara, X. Li, **Y. Park**. **Directional Alignment Mitigates Reward Hacking in Reinforcement Learning for Language Models.** ICML AIWILD workshop.

[11] OPT LLM A. Khaled, K. Ozkara, T. Yu, **Y. Park**†. **MuonBP: Faster Muon Optimizer via Block-Periodic Orthogonalization.** *International Conference on Learning Representations (ICLR)*, 2026.

[12] LLM SYS S. Bian, T. Yu, **Y. Park**†. **Scaling Laws Meet Model Architecture: Toward Inference-Efficient LLMs.** *International Conference on Learning Representations (ICLR)*, 2026. Highlighted at [**Amazon Science Blog**].

[13] LLM RL H. Liu, J. Huang, **Y. Park**, Y. Wang. **Not-a-Bandit: Provably No-Regret Drafter Selection in Speculative Decoding for LLMs.** *International Conference on Learning Representations (ICLR)*, 2026.

[14] ML LLM J. Kim, R. Saha, M. Sung, **Y. Park**. **Demystifying Transition Matching: When and Why It Can Beat Flow Matching.** *International Conference on Artificial Intelligence and Statistics (AISTATS)*,

2026.

[15] LLM OPT Q. Hong, C. Chung, **Y. Park**, M. Hong. **RoSTE: An Efficient Quantization-Aware Supervised Fine-Tuning Approach for Large Language Models.** *International Conference on Machine Learning (ICML)*, 2025.

[16] LLM H. Liu, R. Saha, **Y. Park**, Y. Wang. **ProxSparse: Regularized Learning of Semi-Structured Sparsity Masks for Pretrained LLMs.** *International Conference on Machine Learning (ICML)*, 2025.

[17] TS LLM L. Masserano, A. Ansari, C. Faloutsos, M. Mahoney, A. Wilson, **Y. Park**, Y. Wang. **Enhancing Foundation Models for Time Series Forecasting via Wavelet-based Tokenization.** *International Conference on Machine Learning (ICML)*, 2025.

[18] LLM SYS A. Tseng, T. Yu, **Y. Park**†. **Training LLMs with MXFP4.** *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2025.

[19] LLM OPT K. Ozkara, T. Yu, **Y. Park**†. **Stochastic Rounding for LLM Training: Theory and Practice.** *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2025.

[20] ML B. Kevton, B. Oreshkin, **Y. Park**, R. Song. **Contextual Posterior Sampling with a Diffusion Model Prior.** *Neural Information Processing Systems (NeurIPS)*, 2024.

[21] LLM SYS **Y. Park**†, K. Budhathoki, L. Chen, J. Kübler, J. Huang, M. Kleindessner, J. Huan, V. Cevher, Y. Wang, G. Karypis. **Survey: Inference Optimization of Foundation Models on AI Accelerators.** *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2024.

[22] LLM OPT T. Gautam, **Y. Park**†, H. Zhou, P. Ramen. **Variance-reduced Zeroth-Order Methods for Fine-Tuning LLMs with just Forward Pass.** *International Conference on Machine Learning (ICML)*, 2024.

[23] LLM OPT T. Yu, G. Gupta, K. Gopalswamy, **Y. Park**, J. Huan, R. Diamond. **Collage: Light-Weight Low-Precision Strategy for LLM Training.** *International Conference on Machine Learning (ICML)*, 2024.

[24] LLM UQ C. Marx, W. Ha, C. Bock, J. Huan, **Y. Park**†. **Eliciting Calibrated Uncertainties from Language Models.** *Amazon Machine Learning Conference (AMLC)*, 2024.

[25] ML H. Ding, Y. Ma, **Y. Park**, A. Deoras, H. Wang, B. Kveton. **Trending Now: Modeling Trend Recommendations.** *ACM International Conference on Recommender Systems (RecSys)*, 2023.

[26] ML TS H. Hasson, D. Maddix, Y. Wang, **Y. Park**. **Theoretical Guarantees of Learning Ensembling Strategies with Applications to Time Series Forecasting.** *International Conference on Machine Learning (ICML)*, 2023.

[27] ML TS L. Lui, **Y. Park**†, N. Hoang, H. Hasson, J. Huan. **Robust Multivariate Time-Series Forecasting: Adversarial Attacks and Defense Mechanisms.** *International Conference on Learning Representations (ICLR)*, 2023.

[28] UQ ML C. Marx, **Y. Park**†, H. Hasson, Y. Wang, J. Huan, S. Ermon. **But Are You Sure? An Uncertainty-Aware Perspective on Explainable AI.** *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2023.

[29] TS RL Y. Ding, **Y. Park**†, K. Gopalswamy, Y. Wang, J. Huan. **Dynamic Ensembling for Probabilistic Time Series Forecasting: Reinforcement Learning Approach.** *KDD Time Series MILETS Workshop*, 2023.

[30] ML TS X. Jin, **Y. Park**†, D. Maddix, Y. Wang. **Domain Adaptation for Time Series Forecasting via Attention Sharing.** *International Conference on Machine Learning (ICML)*, 2022.

[31] TS UQ **Y. Park**†, D. Maddix, J. Gasthaus, Y. Wang. **Learning Quantile Functions without Quantile Crossing for Distribution-free Time Series Forecasting.** *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2022.

[32] ML TS UQ T. Yoon, **Y. Park**†, E. Ryu, Y. Wang. **Robust Probabilistic Forecasting via Randomized Smoothing.** *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2022.

[33] TS UQ K. Kan, F. Aubet, T. Januschowski, **Y. Park**, K. Bendis, J. Gasthaus. **Multivariate Quantile**

Functions for Forecasting. *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2022. **Selected as oral, 2.6% of all submissions.**

[34] TS ML L. Masserano, S. Rangapuram, R. Nirwan, S. Kapoor, **Y. Park**, M. Bohlke-Schneider. **Adaptive Sampling for Probabilistic Forecasting Under Distribution Shift.** *NeurIPS Workshop on Distribution Shift*, 2022.

[35] TS LLM X. Zhang, X. Jin, K. Gopalswamy, **Y. Park**, D. Maddix, Y. Wang. **First De-Trend then Attend: Rethinking Attention for Time-Series Forecasting.** *NeurIPS Workshop on Attention Models*, 2022.

[36] TS J. Zhang, **Y. Park**, H. Hasson, D. Maddix, D. Roth, Y. Wang. **Reverse Causal Inference on Panel Data via Generalized Synthetic Control.** *NeurIPS Workshop on Causal Dynamic System*, 2022.

[37] TS A. Jambulapati, H. Hasson, **Y. Park**, Y. Wang. **Testing Causality of High-Dimensional Data.** Available at ArXiv, 2022.

[38] TS **Y. Park**. **On the Explainability of Deep Forecasting Models.** *Amazon Machine Learning Conference (AMLC)*, 2021.

[39] OPT TS Y. Lu, **Y. Park**†, L. Cheng, Y. Wang, D. Foster. **Variance Reduced Training with Stratified Sampling for Forecasting Models.** *International Conference on Machine Learning (ICML)*, 2021.

[40] OPT RL **Y. Park**, R. Rossi, Z. Wen, G. Wu, H. Zhao. **Structured Policy Iteration for Linear Quadratic Regulator.** *International Conference on Machine Learning (ICML)*, 2020.

[41] OPT J. Kim, **Y. Park**, J. Fox, S. Boyd, W. Dally. **Optimal Operation of a Plug-in Hybrid Vehicle with Battery Thermal and Degradation Model.** *American Control Conference (ACC)*, 2020.

[42] OPT **Y. Park**, S. Dhar, S. Boyd, M. Shah. **Variable Metric Proximal Gradient Method with Diagonal Barzilai-Borwien Stepsize.** *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2020.

[43] OPT **Y. Park**, E. K. Ryu. **Linear Convergence of Cyclic SAGA.** *Optimization Letters*, 2020.

[44] OPT RL **Y. Park**, K. Mahadik, R. Rossi, G. Wu, H. Zhao. **Linear Quadratic Regulator for Resource-Efficient Cloud Services.** *ACM Symposium on Cloud Computing (SOCC)*, 2019.

[45] OPT ML **Y. Park**, D. Hallac, S. Boyd, J. Leskovec. **Learning the Network Structure of Heterogeneous Data via Pairwise Exponential Markov Random Fields.** *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2017.

[46] OPT ML D. Hallac, **Y. Park**, S. Boyd, J. Leskovec. **Inferring Time Varying Networks via Graphical Lasso.** *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2017.

†Corresponding author

Invited Talks (Selected)

2026	ML Joint Workshop, KAIST AI, Seoul, S. Korea
2026	K-PAI, Santa Clara, USA
2026	Canon USA Research, Virtual
2025	King Abdullah University of Science and Technology (KAUST), Saudi Arabia
2023	University of California, San Diego
2022	Amazon Machine Learning Conference (AMLC), Virtual
2022	AWS AI Labs, Santa Clara
2021	Amazon Machine Learning Conference, Virtual
2020	Seoul National University, South Korea
2020	Amazon Web Service (AWS) AI, Palo Alto
2020	Facebook AI Research (FAIR), Menlo Park
2020	Rakuten Research, San Mateo
2019	Adobe Research, San Jose
2019	Hyundai AI Labs, Seoul, Korea
2018	Hyundai Global Forum, San Diego
2017	Kakao Brain, Bundang, Korea
2017	Bosch AI, Palo Alto

Honors & Awards

- Top 10 Most Read Blog, Amazon Science, Dec. 2022
- Top 3 Most Read Paper, Amazon Science, Jun. 2022
- Best Presenter Award in AI Session, Hyundai Global Forum, 2018
- Kwanjeong Graduate Fellowship ($110,000 over 2 years), 2013–2015
- Fulbright Graduate Fellowship (Declined), 2013
- National Science and Engineering Scholarship, KOSAF, 2006–2009

Patents

- Model Explainability Insight for Time Series Forecasting (US Patent, 2021)
- System and Method for Resource Scaling for Efficient Resource Management (US Patent, 2020)

Open-Source Contributions

- **NxDT** (Neuron Distributed Training): Foundation model training framework for AWS Trainium
  https://awsdocs-neuron.readthedocs-hosted.com/
- **GluonTS**: Probabilistic time series modeling in Python (1.5K+ GitHub stars)
  https://github.com/awslabs/gluon-ts
- **SnapVX**: Convex optimization solver for problems on graphs
  http://snap.stanford.edu/snapvx/
- **TVGL**: Time-varying graphical lasso for network inference
  https://github.com/davidhallac/TVGL
- **PEMRF**: Graphical structure inference via pairwise exponential MRF
  https://github.com/youngsuk0723/PE-MRF-Code

Academic Collaborators

Stanford: Stephen P. Boyd, Jure Leskovec, Tsachy Weissman, Michael Saunders
Other Academia: Ernest K. Ryu (Seoul National U), Mingyi Hong (U Minnesota), Volkan Cevher (EPFL), Yuxiang Wang (UCSD), Hongseok Namkoong (Columbia)
Industry Research: George Karypis (AWS), Luke Huan (AWS), Yida Wang (AWS), Yuyang Wang (AWS), Dean

Foster (Amazon), Branislav Kveton (Adobe Research), Zheng Wen (Google DeepMind), Suju Rajan (LinkedIn), Mohak Shah (LG Electronics)

Professional Service

Conference Reviewing: NeurIPS, ICML, ICLR, AISTATS, KDD
Journal Reviewing: JMLR, TMLR, IEEE TPAMI, SIAM Journal on Mathematics of Data Science

Research Mentorship

- **Naomi Sagan**, Stanford EE (current intern)
- **Sungyoon Kim**, Stanford EE (current intern)
- **Stanley Wei**, Princeton CS (current intern)
- **Yushu Li,** University of British Columbia CS (current intern)
- **Justin Kim**, SNU EECS (current intern)
- **Wenlong Deng**, University of British Columbia CS → Meta MSL
- **Jason Yoo**, University of British Columbia CS→ AWS AI
- **Jiin Woo**, CMU Computer Science → AWS AI
- **Ahmed Khaled Ragab**, Princeton EECS → Google DeepMind
- **Song Bian**, UW-Madison Computer Science → Nvidia AI
- **Jaihoon Kim**, KAIST Computer Science
- **Albert Tseng**, Cornell Computer Science → OpenAI
- **Kaan Ozkara**, UCLA Electrical Engineering → AWS AI
- **Licong Lin**, UC Berkeley Statistics
- **Chung Yiu Yau**, Chinese University of Hong Kong Computer Science
- **Hongyi Liu**, Rice Computer Science → AWS AI
- **Tanmay Gautam**, UC Berkeley EECS → Microsoft
- **Shingen Sun**, Northwestern Applied Mathematics
- **Tao Yu**, Cornell Computer Science → AWS AI
- **Charlie Marx**, Stanford Computer Science → Hedge fund
- **Yuhao Ding**, UC Berkeley Operations Research → Hedge fund
- **Linbo Liu**, UCSD Applied Mathematics → AWS AI
- **Sanae Lotfi**, NYU Statistics → Meta FAIR
- **Jiayao Zhang**, UPenn Applied Mathematics → Hedge fund
- **Luca Masserano**, CMU Statistics → Meta
- **Xiyuan Zhang**, UCSD Computer Science → AWS AI
- **Arun Jambulapati**, Stanford Mathematics → Postdoc
- **Kelvin Kan**, Emory Mathematics → Postdoc
- **Shantnu Gupta**, CMU Computer Science
- **Taeho Yoon**, Seoul National University Mathematics → Postdoc
- **Yucheng Lu**, Cornell Computer Science → TogetherAI
- **Xiaoyong Jin**, UCSB Computer Science → AWS AI